

3%, and our pooled difference-in-differences estimates rule out effects larger than 2%.¹¹ Second, these effects show no differential trends after the introduction of AI chatbots, indicating that the minimal impacts are not merely a very short-term phenomenon. Lastly, despite evidence in Section 1.4 that employer encouragement boosts work-related benefits of AI chatbots, we find no differential changes in earnings for workers whose employers encourage their chatbot use.

Heterogeneity by Strength of Treatment. Figure 4, Panel (a) examines the earnings impacts of chatbot use by the strength of the adoption treatment.

The top three rows consider the effects of additional investments in employer-provided chatbots and training. These estimates offer insights into the impacts of AI chatbots when firms pursue a more comprehensive strategy. Yet even when the “full package” of encouraged use, enterprise chatbots, and training substantially increases reported benefits (Section 1.4), we find no detectable impact on adopter earnings.

The next two rows focus on more intensive users—those who engage with AI chatbots daily or who adopted early (within the first year after ChatGPT’s launch). These workers have not experienced earnings gains either.

The remaining rows examine workers who report greater benefits from AI chatbot usage: time savings exceeding one hour per day, higher-quality output, enhanced creativity, and engagement with new AI-related tasks. While these self-reported benefits are substantial, the estimates show none of them is associated with better labor market outcomes in our administrative data.

Taken together, Figure 4.(a) suggests that the null earnings effects are not simply the result of weak treatments—at either the firm or worker level. Workers in firms that adopt a more active, multifaceted approach to AI chatbots do not see differential earnings impacts. Nor do those who use the tools more frequently, adopted them earlier, or report

¹¹The dynamic difference-in-differences in Figure 2.(b) reveal that adopters are on slightly stronger labor market trends. However, because these trends entirely predate AI chatbots, the pooled difference-in-differences in Figure 5.(a) (which control for pre-trends; see Equation (2)) are precise zeros, with confidence bands ruling out effects larger than 2%.

large work-related benefits.

Occupational Heterogeneity. The overall zero effects could mask important heterogeneity across occupations. For instance, since teachers are covered by collective bargaining with centralized wage setting, it may be less surprising that individual productivity gains do not translate into higher earnings. By contrast, occupations such as marketing and software development feature decentralized wage setting, allowing more flexibility to adjust pay based on individual productivity. In Appendix C.1, we provide an overview of the wage-setting systems in each occupation. Comparing across occupations thus provides a way to assess whether the limited impacts of AI chatbots reflect labor market rigidities.¹²

Figure 5, Panel (a) shows the estimated effects of AI chatbots on worker earnings for each of the 11 occupations in our sample. We find no significant effects in any occupation: the point estimates are all near zero and the confidence bands generally rule out effects larger than 10%. Even in our two most contrasting occupation groups—software developers and marketers (with flexible wages) versus teachers (with centralized wage-setting)—we see no systematic differences in impact.

Discussion. Our administrative data show that users of AI chatbots have not experienced better outcomes in the labor market. Even among workers who use the tools daily and report substantial benefits, we find no net changes in earnings or hours, not even a break in trend.

Are the weak wage effects of chatbot use theoretically surprising? Competitive labor markets (e.g., our framework in Appendix A.2) imply that workers should be compensated for productivity gains that are not driven by employer initiatives (Acemoglu and Pischke, 1998; Becker, 1964). From this perspective, it is surprising that individual-specific benefits from AI chatbots—such as those among non-encouraged or high-use, high-benefit adopters—do not translate into higher earnings. On the other hand, if employers do

¹²Furthermore, marketing professionals and software developers are exactly the occupations with the highest reported use and productivity gains from the tools (Appendix D.3).

not endorse or invest in the tools, they may be reluctant to reward their use. From this perspective, it is surprising that adopters supported by employer initiatives have not experienced earnings gains.

Another possibility is that workers simply overestimate the actual benefits they derive from the tools (Becker et al., 2025; Edelman, Ngwe and Peng, 2023).¹³ The key contribution of our study is to link these responses to third-party recorded administrative data, allowing us to move the analysis beyond self-reported effects.

3.2 Occupational Mobility

Figure 2, Panel (c) shows that adopters have seen greater occupational switching since the arrival of AI chatbots. By 2024, the additional hours worked in their latest occupations amount to about 4% of a full-time equivalent. The figure provides four insights.

First, the flat pre-trend prior to the arrival of chatbots suggests that the association between occupation switching and chatbot adoption reflects a real effect rather than omitted confounders (e.g., more dynamic workers both adopting chatbots and switching jobs more generally). This contrasts with the earnings gaps in Panel (a), which were entirely attributable to type-specific selection.

Second, the figure supports the notion that chatbots appeal to workers who are new to their occupations. This correlation may reflect that chatbots facilitate occupational switching or, perhaps more likely, that they are simply more attractive after a switch. While the difference-in-differences estimates do not settle the direction of causality, they clearly establish that chatbot adoption and occupational switching are intrinsically linked.¹⁴

Third, the significant estimates provide reassurance that our empirical specification can detect effects where one might hold a prior that a dynamic correlation exists, offering a useful contrast to other outcomes (e.g., worker earnings or workplace employment)

¹³Consistent with experimental evidence from Becker et al. (2025), Appendix E.1.6 shows that workers who report greater time savings from chatbots are also more likely to have new tasks from the tools.

¹⁴In contrast, Appendix E.2.2 shows that the association between chatbot adoption and workplace mobility is weaker, and occupational mobility is no stronger at workplaces that encourage chatbot use. These patterns align with Figure 4, Panel (b), which indicates that occupational switching is driven primarily by individual rather than firm-level factors.

where we have weaker priors about the direction or magnitude of effects.

Finally, although the patterns in occupational mobility are clear and statistically significant, the estimated effects are modest—amounting to roughly a 4% increase in full-time equivalent employment—and not large enough to generate net gains in hours or earnings for adopters (Section 3.1).

Heterogeneity by Strength of Treatment. Figure 4, Panel (b) examines how the relationship between chatbot adoption and occupational mobility varies with the strength of the adoption treatment. The figure shows that the association between chatbot adoption and occupational switching is driven primarily by individual-level factors rather than firm-level initiatives. For example, our main estimate on occupational switching rises from 1.4% to 3.0% for individuals who use the tools daily, and to 3.8% for those who report large time savings.¹⁵ By contrast, the relationship between adoption and occupation switching is fairly similar across employer chatbot initiatives, ranging in 2024 from about 1.6% for non-encouraged adopters to about 0.9% for those with both encouraged-use, enterprise chatbots, and training.

Occupational Heterogeneity. Figure 5, Panel (c), splits the effects of chatbot adoption on mobility by occupation. Positive effects appear in multiple occupations, but are strongest and most statistically significant among IT supporters, and to some extent, office clerks. These are jobs where individual workers likely have more flexibility in choosing their toolset, allowing newcomers to take advantage of AI chatbots. In contrast, financial advisors and customer support specialists often work with established IT systems, making it harder for newcomers to compensate for missing skills by using chatbots on an individual basis. These patterns align with Dillon et al. (2025), who show that workers primarily adopt Microsoft Copilot for tasks where they can use the tools without coordinating with others. Finally, some occupations, such as teachers and accountants, require licensing

¹⁵The effect magnitudes in Figure 4, Panel (b), are generally smaller than the endline estimates in Figure 2, Panel (c), reflecting that the pooled specifications in Figure 4 control for time trends (see Equation (2)).